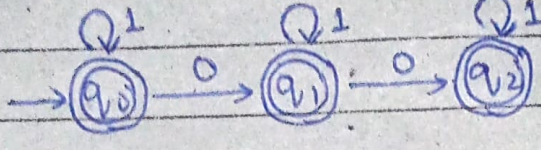
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Regular Expressions

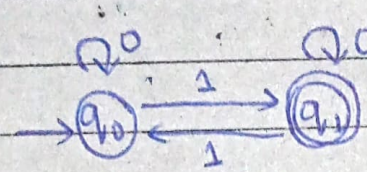
Expressions which describe the language accepted by a DFA.

Q design a DFA over $\Sigma = \{0,1\}$ such that each string has atmost two zeroes.



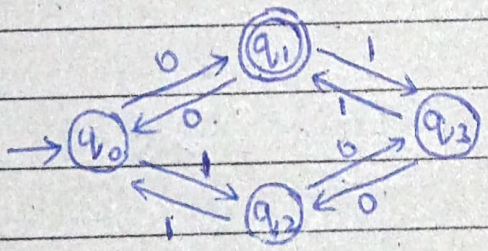
$L = \{\epsilon, 1, 11, 110, 1101, 11011, 110110, 1101101, 11011011, \ldots\}$

Q design a DFA over $\Sigma = \{0,1\}$ such that each string has odd number of 1s.



$L = \{0, 00, 001, 10, 100, 111, 1011, 1101, 10101, 110101, \ldots\}$

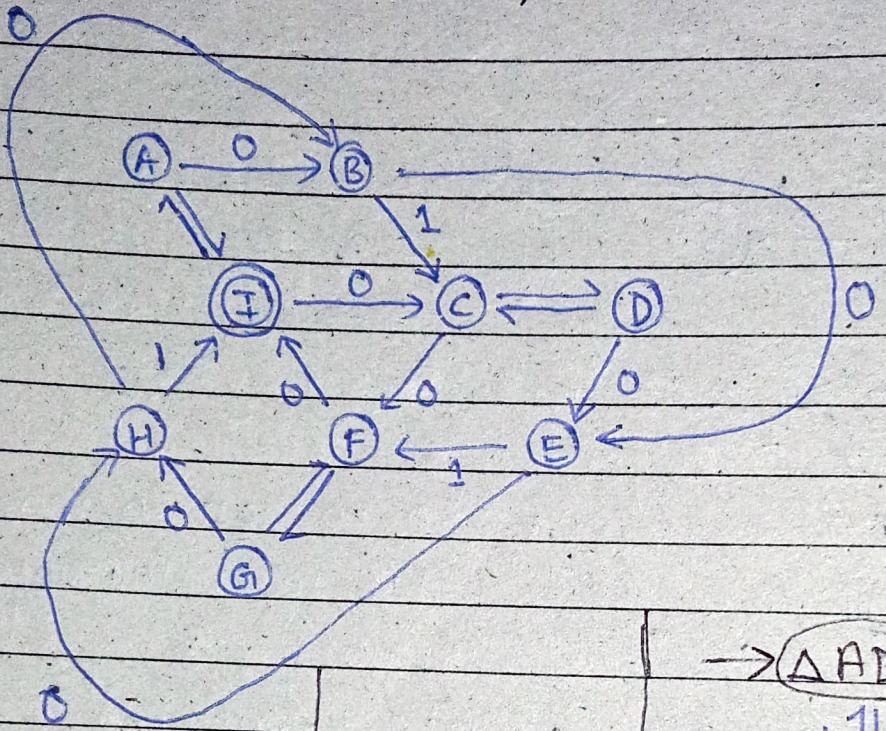
Q design a DFA over $\Sigma = \{0,1\}$ such that each string has odd no. of 0's and even no. of 1's.



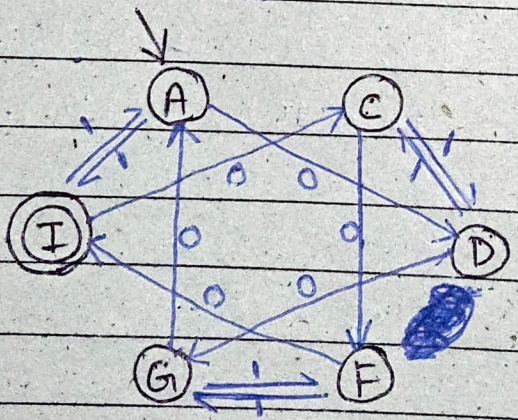
$L = \{0, 000, 011, 01100, 110, 101, 10010, \ldots\}$

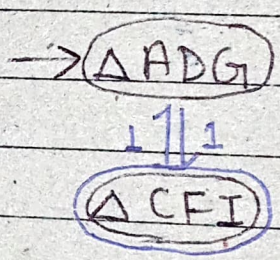
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Q design a DFA over $\Sigma = \{0,1\}$, such that each string has count(1) = 2m+1 , count(0) = 3n

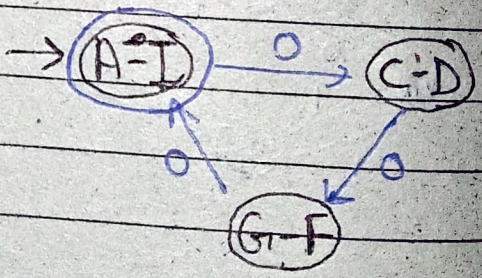


minimisation





count(1) = 2m+1



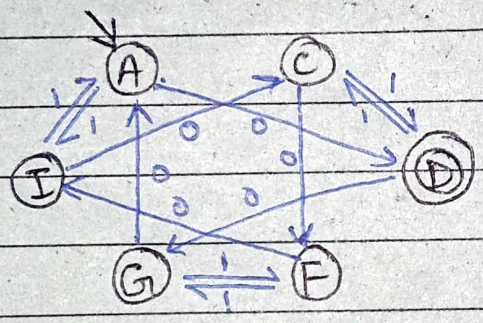
count(0) = 3n

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generalisation : let, x = count(1)%2 & y = count(0)%3
then we can design an automata for different values of x & y by just changing the final state accordingly.

y \ x	0	1
0	A	I
1	D	C
2	G	F

Q design a DFA over $\Sigma = \{0,1\}$, where
count(1) = 2m & count(0) = 3m+1
we have x = 0, y = 1
⇒ final state, F = {D}



Q design a DFA, which accepts all binary strings whose decimal equivalent are multiple of 5.

we know, string · 0 = string * 2 + 0
string · 1 = string * 2 + 1

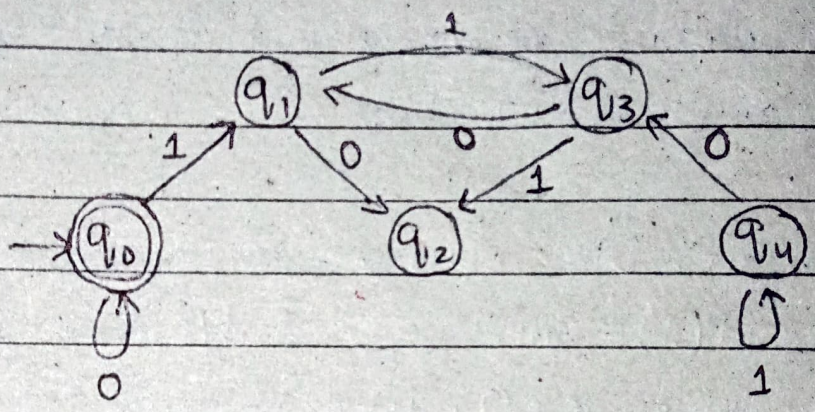
Let
x = string%5
y = (string · 0)%5 ⇒
z = (string · 1)%5

x	y	z
0	0	1
1	2	3
2	4	0
3	1	2
4	3	4

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hence the corresponding transition table, and DFA will look like.

	0	1
→ q_0	0	1
q_1	2	3
q_2	4	0
q_3	1	2
q_4	3	4

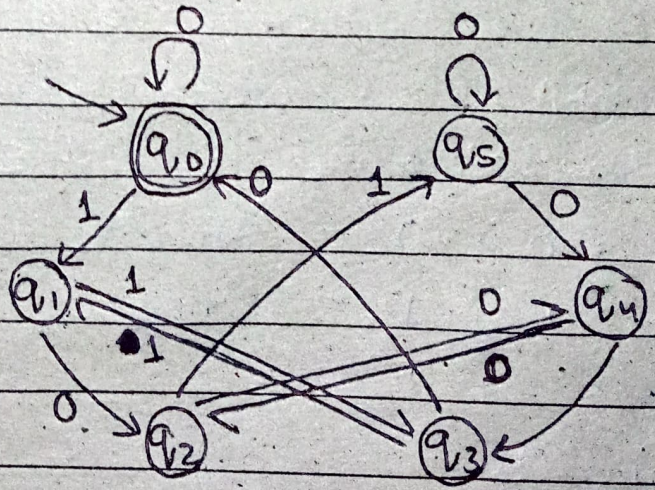


since we are interested in the strings which are multiple of 5, i.e, string%5=0. Hence we will q_0 the final state.

Q design a DFA, which accepts all binary strings whose decimal equivalent are multiple of 6

Applying the same logic as in the ~~abo~~ prev. question, we get.

	0	1
→ q_0	0	1
q_1	2	3
q_2	4	5
q_3	0	1
q_4	2	3
q_5	4	5



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Regular Expression Identities

$R + R = R$

$R \cdot R \neq R$

$R + P = P + R$

$R \cdot P \neq P \cdot R$

$R + \Lambda \neq R / \Lambda$

$R \cdot \Lambda = R = \Lambda \cdot R$

$R + \phi = R = \phi + R$

$R \cdot \phi = \phi = \phi \cdot R$

$(R^*)^* = R$

$R \cdot R^* = R^+ = R^* \cdot R$

$\Lambda + R^+ = R^* = R^+ + \Lambda$

$P(QP)^* = (PQ)^* P$

$(P+Q)^* = (P^* Q^*)^* = (P^* + Q^*)^*$

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Arden's Theorem

It states that if we have $R = Q + RP$ and then we can also write it as $R = QP^*$

proof: we have $R = Q + RP$ —①

sub ① in ①, we get,

$R = Q + (Q+RP)P$

$R = Q + QP + RPP$ —②

sub ① in ② we get,

$R = Q + QP + (Q+RP)PP$

$R = Q + QP + QPP + RPP$

after n^{th} substitution, we will get

$R = Q + QP + \ldots + QP^n + RP^{n+1}$

$R = Q(1 + P + \ldots + P^n) + RP^{n+1}$

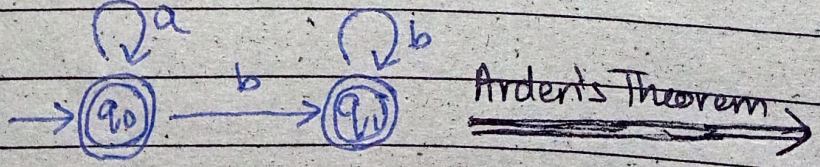
$R = QP^n + RP^{n+1}$ (where $n \geq 0$)

Ignoring the RP^{n+1} term, we get

$R = QP^*$

Arden's theorem is used to convert given finite automata to regular expression

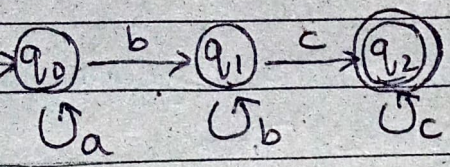
eg:



Arden's Theorem ⟹ a^*b^*

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Q Find the regular expressions for the following automata.

(i) 

we have, $q_0 = \Lambda + q_0 a$, applying Arden's theorem

A.T. we get, $q_0 = \Lambda a^* = a^*$ ——— (1)

we have, $q_1 = q_0 b + q_1 b$

$q_1 = a^* b + q_1 b \quad [\because q_0 = a^*]$

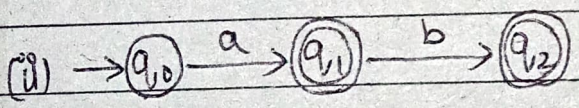
A.T. we get, $q_1 = (a^* b) b^* = a^* b b^*$ ——— (2)

we have, $q_2 = q_1 c + q_2 c$

$q_2 = (a^* b b^*) c + q_2 c \quad [\because q_1 = a^* b b^*]$

A.T. we get, $q_2 = (a^* b b^* c) c^*$

$\Rightarrow$ $\underline{q_2 = a^* b b^* c c^*}$

(ii) 

we have, $q_0 = \Lambda$

we have, $q_1 = q_0 a = \Lambda a = a \quad [\because q_0 = \Lambda]$

we have, $q_2 = q_1 b = ab \quad [\because q_1 = a]$.

Reg. Exp. $= q_1 + q_2$

$= \underline{a + ab}$.

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Pumping Lemma (proof by contradiction)

- To prove: L is irregular, we assume it to be regular and it has an FA with 'k' no. of states. Then,
  - → select $w \in L$ such that $|w| \geq k$
  - → divide w in x.y.z. such that
    - → $|xy| \leq k$
    - → $y \neq \varepsilon$
    - → $\forall\, i \geq 0$, $xy^iz \in L$

eg: $L = a^n b^n \;\forall\, n \geq 0$

let its FA has k states.

let $w = a^k b^k$ ∵ $w \in L$ & $|a^k b^k| = 2k > k$

also

$w = \underbrace{a^{k-p}}_{x} \cdot \underbrace{a^p}_{y} \cdot \underbrace{b^k}_{z} \quad [0 < p \leq k]$

now for $i = 0$,

$xy^iz = a^{k-p} \cdot (a^p)^0 \cdot b^k = a^{k-p} b^k \notin L$

Hence this language is irregular.

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eg: $L = a^n b^m \quad \forall\ m,n \geq 0$

Let its FA has k states

Let $w = a^k b^k$ $\because a^k b^k \in L$ & $|a^k b^k| = 2k > k$

also

$w = \underbrace{a^{k-p}}_{x} \cdot \underbrace{a^p}_{y} \cdot \underbrace{b^k}_{z}$

i	xy^iz	$xy^iz \in L$
0	$a^{k-p}b^k$	✓
1	$a^k b^k$	✓
2	$a^{k+p} \cdot b^k$	✓
3	$a^{k+2p} \cdot b^k$	✓
⋮	⋮	⋮

eg: $L = a^n b^m c^{n+m} \quad \forall\ m,n \geq 0$

Let its FA has k states

Let $w = a^k b^k c^{2k}$ $\because a^k b^k c^{2k} \in L$ & $|a^k b^k c^{2k}| = 4k$

also,

$w = \underbrace{a^{k-p}}_{x} \cdot \underbrace{a^p}_{y} \cdot \underbrace{b^k c^{2k}}_{z} \quad [0 < p \leq k]$

now for $i = 0$,

$xy^iz = a^{k-p} \cdot (a^p)^0 \cdot b^k c^{2k}$

$= a^{k-p} \cdot b^k c^{2k} \notin L \quad [\because (k-p)+k \neq 2k]$

Hence this language is irregular.